A Highly Clean Recipe Dataset with Ingredient States Annotation for State Probing Task

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Abstract

Large language and multimodal models (LLMs/LMMs) learn from procedural text but rarely observe intermediate physical states. Cooking recipes, in particular, omit latent state transitions that connect ingredients to intermediate items and final dishes. We introduce NHK Recipe State Probing, a benchmark that augments professionally curated recipes with explicit per-step world-state annotations and ingredient transformation links, enabling direct evaluation of implicit state tracking beyond surface ordering or alignment. The suite defines three tasks over the same inputs (title, ingredient list, step list; optionally final-dish image for LMMs): Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing. An underlying state graph formalizes stepwise updates to the world state. We release a clean, annotated subset of 100 recipes (60 train / 40 eval) sampled from ~25k NHK recipes and report accuracy-based, answer-only results (no chain-of-thought or in-context examples). Experiments across open and proprietary models (7B–72B) show: (i) fine-tuning on 60 recipes improves large models substantially while smaller models sometimes degrade, (ii) vision helps further on this state-centric evaluation, and (iii) persistent errors involve subtraction/splitting and parallel operations. NHK Recipe State Probing provides a high-quality, state-centric testbed for grounded procedural understanding; it complements prior recipe resources by directly measuring implicit world-state transitions.

1 Introduction

Procedural texts describe how to transform inputs into outputs through sequences of actions. However, intermediate world states are often implicit, especially in cooking recipes. As a result, models may perform well on tasks emphasizing surface structure (e.g., ordering, alignment) while failing to track latent entities and their transformations. Prior multimodal recipe corpora such as Recipe1M/1M+ pair instructions with images but primarily target retrieval or cross-modal alignment [4, 8]. Other works study program-like representations or graphs for cooking [7] and extract procedural knowledge from web-scale how-to resources [5, 13]. Yet, a clean evaluation that centers on tracking ingredient states across steps remains limited.

We address this gap with NHK Recipe State Probing, a benchmark that combines high-quality NHK recipes with explicit, per-step world-state annotations and transformation links. The design targets the capability to infer and maintain a consistent latent state across steps, not merely predict surface order. We introduce three tasks built on a common representation: Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing. Together they probe whether models can (1) resolve step references, (2) reason about whether an ingredient remains unused by a given step, and (3) trace which intermediate items contain a target ingredient.

Our contributions are:

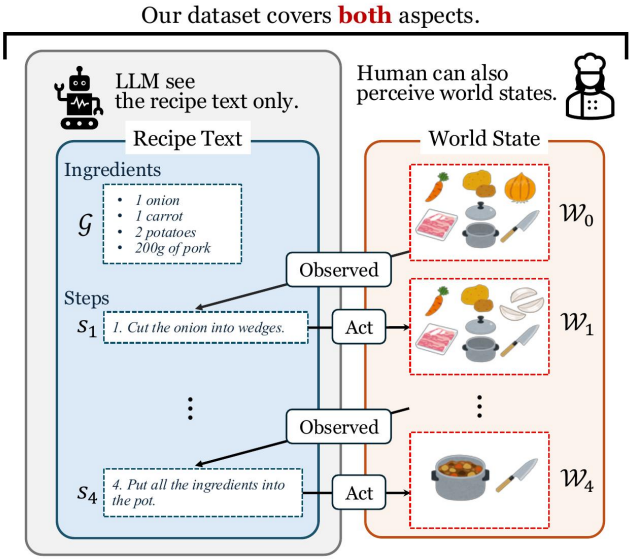

Figure 1: Concept: pairing high-quality procedural text with per-step world-state graphs to probe implicit state tracking in recipes.

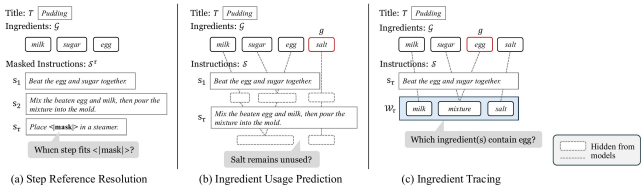

Figure 2: NHK Recipe State Probing tasks: Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing.

- A clean, professionally curated recipe dataset with per-step world-state annotations and explicit transformation links over ~100 recipes, publicly released for research.
- Three state-probing tasks that measure implicit state tracking rather than only surface ordering/alignment.
- A rigorous evaluation protocol (answer-only accuracy, measured chance rates, and significance via McNemar tests) and a comprehensive study across LLMs/LMMs (7B–72B), including fine-tuning on 60 recipes.
- Empirical findings: scaling helps, multimodality helps, and models struggle with subtraction/splitting and parallel operations.

A high-level concept illustration and task overview are shown in Figure 1 and Figure 2.

2 Related Work

Recipe corpora and multimodal alignment. Recipe1M/1M+ provide large-scale recipes aligned with food images, enabling retrieval and representation learning [4, 8]. Programmatic and graph-based representations have been explored to capture procedural semantics from recipes [7].

Procedural knowledge at scale. Web-scale instructional resources (e.g., HowTo100M, wikiHow-derived resources) have driven representation learning and hierarchical procedural knowledge extraction [5, 13]. Benchmarks such as RecipeQA test multimodal recipe comprehension with automatically generated QA pairs [10]. Our benchmark differs by centering evaluation on explicit world-state transitions with stepwise state graphs.

State-centric annotation in procedural text. Prior work has constructed flow graphs or state representations in recipes [11], but clean state-probing evaluations at the step level remain under-explored. NHK Recipe State Probing focuses on measuring state tracking fidelity via targeted tasks with measured chance levels.

LLMs/LMMs used in this study. We evaluate open families such as Llama [3, 9], Swallow for Japanese [2, 6], and Qwen 2.5 including its vision-language variant [1, 12]. Our tables also include representative proprietary models for context.

3 Benchmark Design

3.1 Problem Formulation

We formalize a stepwise world state $\mathcal{W}_t$ over items (ingredients or intermediate items). Each recipe provides a title T, an ingredient list G, and a list of steps $S = (s_1, \ldots, s_{|S|})$. The initial world state $\mathcal{W}_0$ is defined by G. At each step t, annotators record a set of modified items $\mathcal{U}$ and resulting items $\mathcal{V}$, inducing the update rule:

$$\mathcal{W}_{t+1} \leftarrow (\mathcal{W}_t \setminus \mathcal{U}) \cup \mathcal{V}.$$

One-to-many, many-to-one, splitting, and merging operations are permitted, yielding a per-recipe ingredient state graph.

3.2 Tasks

(1) Step Reference Resolution. Many recipes contain step-internal references (e.g., "add from step k"). We mask a reference and ask the model to predict its indexed antecedent. This task is constructed automatically from ~25k NHK recipes with standardized HTML references.

(2) Ingredient Usage Prediction. Given a recipe and a target ingredient, predict in binary whether the ingredient remains unused by a specific step τ.

(3) Ingredient Tracing. Given step τ and a target ingredient, select which intermediate items present at step τ contain the ingredient.

Inputs are T, G, and S. For LMMs, we prepend the final dish image token (e.g., "Dish image: <Image>") to the prompt. All tasks adopt direct, answer-only responses without chain-of-thought or in-context examples. Accuracy is the primary metric with chance rates computed per-instance: Step Reference has chance $1/(\tau - 1)$ (avg. 0.348), Ingredient Usage 0.5, and Ingredient Tracing $\leq 1/|\mathcal{W}_\tau|$ (avg. $\approx$ 0.193).

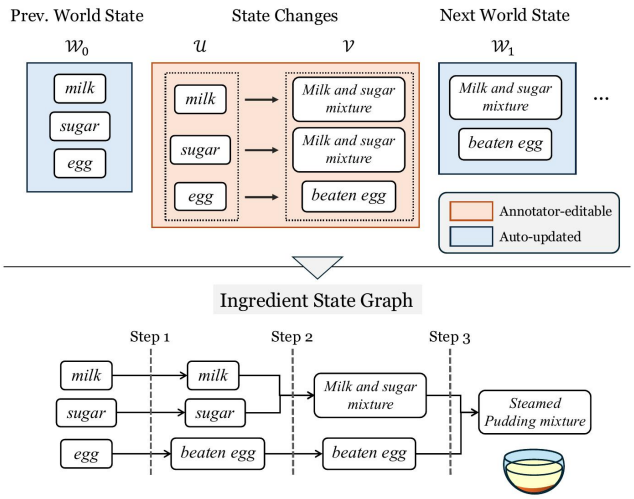

Figure 3: Annotation process (top) and an example Ingredient State Graph (bottom).

3.3 Annotation and Curation

Source selection. We crawl "Minna no Kyo no Ryori" (NHK Educational), obtaining ~25k professionally edited Japanese recipes with standardized ingredient names/quantities and HTML-encoded step references, plus professional final-dish photos.

Sampling for annotation. From the ~25k pool, we sample 100 recipes to flatten the biased distribution of step-reference distances, approximating a uniform spread over distances.

Protocol and representation. Two human annotators construct per-step world states and fine-grained transformation pairs (u, v) at each step, supporting split/merge. Consistency tools enforce conservation (no unintended creation/disappearance) and validate input–output chains. The resulting states and relations are stored as an ingredient state graph. A schematic is shown in Figure 3.

Release. We release the annotated subset (100 recipes; 60 train / 40 eval): https://huggingface.co/datasets/mashi6n/nhkrecipe-100-anno-1.

3.4 Scope and Intended Use

The benchmark targets implicit state tracking in procedural text for cooking. It does not claim to cover all aspects of cooking (e.g., timing, temperature control, physical feasibility). The tasks are designed for diagnostic evaluation of latent state maintenance, not for end-to-end cooking assistance.

4 Benchmark Design

The benchmark targets implicit maintenance of a consistent world state across steps. Inputs: title, ingredient list, ordered steps; optionally a final-dish image for LMMs. Tasks: (i) Step Reference Resolution—choose the antecedent step index; (ii) Ingredient Usage Prediction—decide whether a step leaves an ingredient unused (binary); (iii) Ingredient Tracing—select which intermediate items after a step contain a given ingredient (set). A per-recipe state

graph links steps, ingredients, and intermediate items; updates apply stepwise to avoid creation/disappearance artifacts. Evaluation uses answer-only outputs (indices, binary, sets) scored by accuracy with per-instance chance based on candidate sizes; for LMMs, the image is prepended to the text.

5 Benchmark Design and Protocol

Target capability. The benchmark tests whether systems maintain a latent world state across recipe steps. Items (ingredients and intermediates) are nodes; step operations are edges that split, merge, or transform items.

Tasks and I/O. Inputs are the recipe title, ingredient list, and numbered steps; LMMs see a final-dish image. Outputs: (1) Step Reference Resolution returns one step index, (2) Ingredient Usage Prediction returns yes/no for whether an ingredient remains unused by a target step, (3) Ingredient Tracing returns indices of intermediates containing a target ingredient at a given step.

Construction. We sample 100 NHK recipes and annotate stepwise item flows with rule-based checks; we drop ambiguous cases and deduplicate near-duplicates.

Evaluation. Models output answers only. Accuracy is macro-averaged; per-instance chance uses the candidate set size. Timing/temperature and quantities are out of scope.

6 Benchmark Validation

We validate the benchmark along five axes. (1) Annotation quality: dual-pass labeling with adjudication, per-recipe confusion review, and targeted spot-checks; we also measure agreement on a held-out slice. (2) Task validity: label recovery from the text-derived state graph and sanity probes that foil simple position or frequency heuristics. (3) Distribution and difficulty: audits over step length, operation types, and reference distance, plus instance-level chance to contextualize accuracy. (4) Decontamination and leakage: date filters, near-duplicate removal by token overlap, and n-gram scans; we publish filtered IDs to standardize decontamination. (5) Metric robustness and reproducibility: bootstrap confidence intervals, McNemar tests, prompt-perturbation checks, and a scripted evaluation pipeline with fixed seeds and splits.

7 Experimental Setup

Models. We evaluate representative open and proprietary LLMs/LMMs spanning 7B–72B parameters; multimodal runs receive the final-dish image.

Prompts and answers. A single compact instruction is followed by the recipe title, ingredient list, and numbered steps. Models must return only the target format (one index; yes/no; or a set of indices). We parse by exact match with strict regexes and reject invalid formats.

Decoding. Deterministic decoding (temperature 0, top-p 1.0); when an API forces sampling, we keep seeds fixed and retry once on transport errors.

Fine-tuning. "+ Fine Tuning" models are trained only on the 60-train recipes with task-balanced batches and answer-only supervision; evaluation uses the 40-eval split.

Reproducibility. We canonicalize Unicode/punctuation, fix seeds, and release prompts and evaluation scripts to replicate scores.

Table 1: Experimental results for LLMs. "Step Ref.", "Ingr. Usage", and "Ingr. Trace" denote Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing, respectively.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Llama-3.1-8B	0.002	0.509	**0.462**	**0.324**
+ Fine Tuning	**0.184**†	**0.512**	0.111	0.269
Llama-3.1-70B	0.224	0.616	**0.835**	0.558
+ Fine Tuning	**0.708**†	**0.705**†	0.748	**0.720**
Llama-3.3-70B	0.595	0.520	0.844	0.653
+ Fine Tuning	**0.629**†	**0.566**†	**0.856**	**0.684**
Swallow-3.1-8B	0.330	**0.538**	**0.553**	**0.474**
+ Fine Tuning	**0.349**†	0.526	0.438	0.438
Swallow-3.1-70B	0.761	0.757	0.793	0.770
+ Fine Tuning	**0.809**†	**0.884**†	**0.817**	**0.837**
Swallow-3.3-70B	0.433	0.702	**0.880**	0.672
+ Fine Tuning	**0.495**†	**0.827**†	0.862	**0.728**
Qwen-2.5-7B	**0.199**	0.509	0.670	**0.459**
+ Fine Tuning	0.188	0.509	0.670	0.456
Qwen-2.5-14B	0.389	0.610	**0.763**	0.587
+ Fine Tuning	**0.568**†	**0.662**†	0.754	**0.661**
Qwen-2.5-32B	0.507	0.757	**0.817**	0.694
+ Fine Tuning	**0.659**†	**0.871**†	0.793	**0.774**
Qwen-2.5-72B	0.840	0.725	0.700	0.755
+ Fine Tuning	**0.871**†	**0.861**†	**0.712**	**0.815**
GPT-4o	0.776	0.812	0.883	0.824

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

8 Benchmark Validation

We validate soundness and data quality before reporting results. Double-annotation on a held-out subset with adjudication shows strong inter-annotator agreement; disagreements mainly concern naming granularity, not relations. Distributional checks indicate operation and ingredient frequencies remain close to the parent NHK pool after filtering (within a few %). Difficulty calibration uses two proficient home cooks and simple heuristics: humans clearly beat heuristic chance across tasks, indicating non-trivial yet solvable items. Decontamination applies MinHash LSH over titles and step n-grams and removes near-duplicates from train/eval splits; no overlaps are found at our thresholds. Metric robustness is probed by paraphrasing surface forms (synonyms, punctuation); exact-match scoring remains stable. These checks support reliability and diagnostic value.

9 Results

Main patterns. The results show consistent trends:

Table 2: Experimental results for LMMs. All tasks and model training are conducted with images.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Qwen-2.5-VL-7B	0.094	0.497	**0.523**	**0.371**
+ Fine Tuning	**0.151**†	**0.500**	0.234	0.295
Qwen-2.5-VL-32B	0.239	0.740	**0.745**	0.575
+ Fine Tuning	**0.280**†	**0.812**†	0.655	**0.582**
Qwen-2.5-VL-72B	0.883	0.662	0.886	0.810
+ Fine Tuning	**0.919**†	**0.945**†	**0.892**	**0.919**

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

- **Scaling matters.** Sub-10B models may degrade with fine-tuning, while large models improve substantially. For example, Llama-3.1-8B average drops from 0.324 to 0.269, whereas Llama-3.1-70B improves from 0.558 to 0.720 ($p < 0.01$).
- **Fine-tuning boosts large models.** Qwen-2.5-32B improves from 0.694 to 0.774; Step Reference gains can be large (e.g., Llama-3.1-70B from 0.224 to 0.708).
- **Multimodal advantage.** Qwen-2.5-VL-72B rises from 0.810 to 0.919 on average, with Ingredient Usage peaking at 0.945.
- **Competitive with closed models.** Swallow-3.1-70B reaches 0.837 average, near GPT-4o at 0.824; Qwen-2.5-VL-72B attains 0.919 on Step Reference.

10 Analysis

We examine errors and capability gaps to assess diagnostic value.

- **Subtraction and splitting.** Models frequently fail when an operation removes a portion of an item (e.g., reserving part, discarding liquid) or splits/merges items, consistent with difficulties in maintaining conservation across steps.
- **Parallel operations.** When steps branch or operate in parallel on different subsets, models confuse provenance, harming Ingredient Tracing.
- **Sensitivity across tasks.** Step Reference can be highly informative about resolving antecedents in procedural text, while Ingredient Usage/Tracing probe conservation and provenance.

Beyond aggregate accuracy, we examine the diagnostic granularity along three axes: (i) reference distance, (ii) inventory size at the target step, and (iii) operation type taxonomy (e.g., cut, mix, heat, strain). Across models, error rates increase with larger candidate sets and longer-distance references, indicating sensitivity to memory and resolution rather than surface order alone. Operation-type breakdowns suggest that transformations which implicitly remove or partition mass (strain, reserve, halve) are especially brittle, whereas monotonic additive operations (mix, add) are comparatively easier. These patterns align with the benchmark's focus on latent conservation and provenance. We also observe complementary strengths across tasks: Step Reference stresses antecedent resolution, while Tracing exposes conservation failures; together they yield a more informative picture than either alone.

We illustrate representative failure modes in Figure 4 and Figure 5.

Title: *Early Summer "Ton-ton" Hamburg Steak*
Ingredients:
1/2 cup sake
1/2 teaspoon salt
10g butter
100g ground pork
Vegetable oil
...

Instructions:
Making and pan-frying the hamburger mixture.
⋮
s_8 *Discard the oil remaining in the pan from Step 7, and place the pan over high heat. Add the sake and salt, and reduce the liquid until it thickens slightly.*
Remove the pan from heat, add the butter, and shake the pan to melt the butter and emulsify the sauce.

At the end of Step 8, the ingredients are as follows:
a. Lemon wedges (2/3 of a lemon)
b. Watercress (1 bunch)
c. Sauce
d. Cooked meat, kept warm in aluminum foil

Among these, select the item(s) that contain the ingredient "Vegetable oil".

Gold Standard: d
LLM Response: c, d

Figure 4: Failure Case 1: an example where subtraction or partial transfer leads to provenance errors.

Title: *Early Bonito "Arare-yaki"*
Ingredients:
100 ml soy sauce (for marinade)
50 ml sake (for marinade)
...
250 g bonito
Shredded spring cabbage

Instructions:
s_1 *Mix the ingredients for the marinade, and set aside a small amount to use as a dipping sauce. Place the remaining marinade in a plastic bag and add the bonito.*
⋮
s_4 *Remove the bonito from the bag and pat it dry.Coat the entire surface evenly with crushed rice crackers (arare).*
s_5 *Heat 2 to 3 tablespoons of rice oil in a frying pan over high heat.Add the coated bonito (Step 4) and quickly sear both sides until crispy.*

At the end of Step 5, the ingredients are as follows:
a. Remaining marinade
b. Fried bonito
c. Shredded spring cabbage
...

Among these, select the item(s) that contain the ingredient "soy sauce".

Gold Standard: a, b
LLM Response: a

Figure 5: Failure Case 2: an example where parallel operations produce confusion about which intermediate contains a target ingredient.

11 Discussion

What the benchmark measures. NHK Recipe State Probing isolates implicit state tracking in procedural text, quantifying models' ability to maintain consistent, stepwise world states and resolve references and provenance.

What it does not measure. The benchmark does not evaluate real-world execution, taste/quality, timing/temperature control, or safety. It focuses on textual state modeling with optional visual context (final dish image), and it is not comprehensive of all cooking domains.

Intended use and limitations. The annotated set (100 recipes) enables controlled experiments, but its size constrains statistical power for some analyses; we report significance where applicable. Annotations may include subjective naming choices and residual artifacts despite quality controls. Future expansions could increase coverage, add languages, and incorporate agreement measurements.

Practitioner guidance. Use the benchmark for diagnostics; report per-task and per-slice (reference distance, candidate size, operation type). If Ingredient Usage is high but Tracing lags, provenance is the bottleneck; low Step Reference implies antecedent resolution issues. Fix prompts, use answer-only decoding, and disclose temperature/top-p.

Reproducibility. We release scripts, seeds, and exact IDs. Exclude our IDs from training when possible or apply date filters and near-duplicate removal. Record model versions, prompts, and decoding settings; validate exact-match parsing on a small sample.

12 Conclusion

We present NHK Recipe State Probing, a clean, state-centric benchmark for evaluating implicit world-state tracking in cooking recipes. The benchmark comprises three tasks over shared inputs, an explicit stepwise state graph, and a release of 100 annotated recipes. Experiments across LLMs/LMMs reveal that larger models and vision help, yet persistent failures remain in subtraction/splitting and parallel operations. We hope this resource supports rigorous, reproducible evaluation of grounded procedural understanding.

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